

Experiment 2 — Implementation of Object Storage using Amazon S3

Step 1: Access Amazon S3

1. Log in to the **AWS Management Console**
2. In the top search bar, type **S3**
3. Click on **Amazon S3** — the S3 dashboard opens showing your buckets, storage usage, and management options

 **Screenshot needed:** S3 dashboard/landing page

Step 2: Create an S3 Bucket

1. Click **Create bucket**
2. Enter a **globally unique bucket name** (e.g., yourname-123-bucket)
3. Region: **Asia Pacific (Mumbai) — ap-south-1**
4. Under **Object Ownership** — keep **ACLs disabled (recommended)**
5. Under **Block Public Access** — keep **Block all public access** checked
6. **Bucket Versioning** — leave disabled
7. **Default encryption** — leave as SSE-S3 (enabled by default)
8. Click **Create bucket**

 **Screenshot needed:** Bucket creation config screen, and the success confirmation showing your bucket in the bucket list

Step 3: Upload Objects

1. Click on your newly created bucket name to open it
2. Click **Upload**
3. Click **Add files** and select:
 - One **text file** (.txt)
 - One **PDF or image file** (.pdf / .jpg / .png)
4. Leave all other settings as default
5. Click **Upload** and wait for the success status

 **Screenshot needed:** Upload status screen showing both files as "Succeeded", and the bucket objects view showing both files listed

Step 4: Configure Access Permissions

1. Click on any one of the uploaded objects
2. Go to its **Permissions** tab
3. Observe the following sections and note them down:

Bucket Policy: Written in JSON, defines allowed/denied actions on buckets and objects. By default no policy is set, meaning access is denied unless explicitly granted.

Object Ownership: Set to "Bucket owner enforced" — meaning ACLs are disabled and access is controlled only through policies.

Access Control List (ACL): Since ACLs are disabled, this section shows the bucket owner enforced message. ACLs are a legacy permission system; AWS now recommends using bucket policies instead.

Block Public Access: All four settings are ON, meaning no public access is possible through ACLs or bucket policies.

 **Screenshot needed:** Permissions tab of the bucket showing all four sections — Block Public Access, Bucket Policy, Object Ownership, and ACL

Summary Table — ACL vs Bucket Policy

Include this table in your journal:

Feature	Bucket Policy	ACL
Format	JSON-based	Predefined grants
Granularity	Bucket + Object level	Per object
Flexibility	High	Limited
AWS Recommendation	Preferred	Legacy, discouraged
Public Access Control	Via Block Public Access	Via public ACL grants

Final Summary:

- Open AWS Console and go to S3
- Create an S3 bucket with all default configurations
- Upload a pdf and an image
- Click on them to view the permissions